Decision Rules

Fraud Alert Demo Decision Rules

Purpose

These rules help the demo agent convert an alert plus compact customer history into an explainable fraud confidence level and a recommended next step.

Core rule set

Rule 1: Impossible travel is the strongest signal If the alert shows distant locations in a very short period of time, strongly increase fraud confidence.

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Use this especially when:
- the travel speed is clearly impossible
- the second location is in a different country
- the second location is not present in recent customer history

Recommended outcome:
- high confidence fraud
- escalate for fraud review
- create or update a case
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Rule 2: New geography increases confidence If the alert location is outside the customer's recent pattern, increase confidence.

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Use this when:
- the country is new
- the city is far from recent activity
- the customer history is mostly concentrated in one region

Recommended outcome:
- medium or high confidence depending on other signals
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Rule 3: Amount far above normal increases confidence If the transaction amount is much larger than the customer's recent baseline, increase confidence.

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Use this when:
- recent transactions are small or moderate
- the current amount is significantly larger
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- the merchant is also unfamiliar

Recommended outcome:
- use as supporting evidence, not the only reason
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Rule 4: Familiar behavior lowers confidence If the alert matches the customer's recent history, lower confidence.

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Use this when:
- the city or country has recent matches
- the amount is within the recent range
- the merchant or merchant type is familiar

Recommended outcome:
- low confidence
- likely false positive or monitor only
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Rule 5: Missing context means safe escalation If the history is too limited or key facts are missing, do not over-claim confidence.

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Use this when:
- the transaction history is too small
- the alert is missing important data
- the evidence is mixed

Recommended outcome:
- medium confidence at most
- escalate for human review
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Simple demo scoring guidance

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Use a lightweight explainable approach:
- start from the alert itself
- increase confidence for impossible travel
- increase confidence for new geography
- increase confidence for unusual amount
- increase confidence for unfamiliar merchant behavior
- reduce confidence for familiar history
- reduce confidence when strong evidence is missing
```

Confidence labels

High confidence fraud Use when the alert clearly breaks recent history and includes impossible travel or multiple strong mismatches.

Medium confidence suspicious activity Use when the alert is concerning but the evidence is partial or history is limited.

Low confidence or likely false positive Use when the alert mostly fits recent customer behavior.

Next-step guidance

For high confidence fraud - create or update a case - notify an analyst - recommend immediate review

For medium confidence suspicious activity - create a case or queue item - explain what is suspicious - recommend human review if needed

For low confidence - document why confidence is low - recommend monitor only or no disruptive action

Example decisions

Example 1 Recent history: - Miami hotel purchase

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Alert:
- Tokyo luxury store 10 minutes later

Decision:
- high confidence fraud because the geography is new, the speed is impossible, and the amount is much la
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Example 2 Recent history: - New York coffee shop and bakery

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Alert:
- another small local New York purchase

Decision:
- low confidence because the alert matches the recent behavior pattern
```